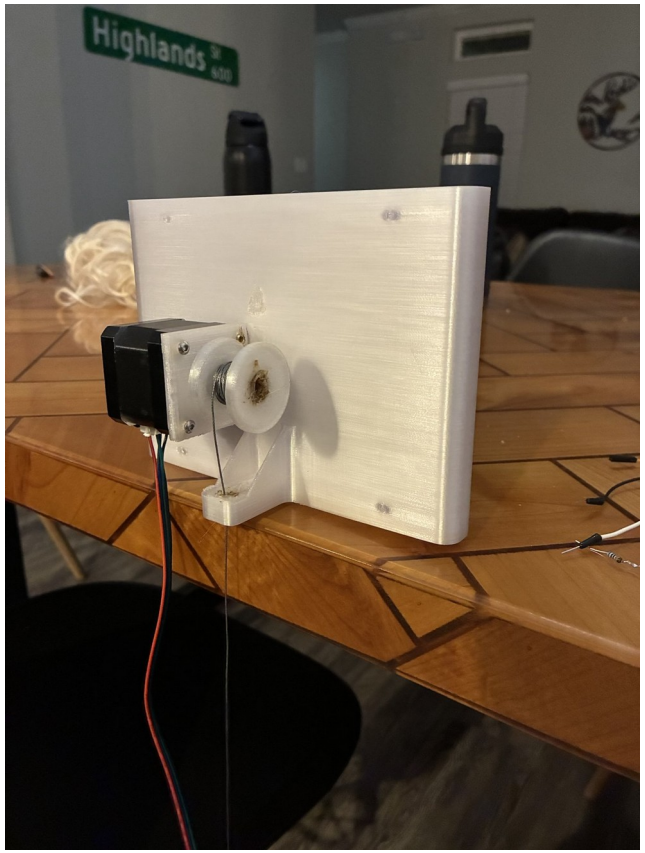
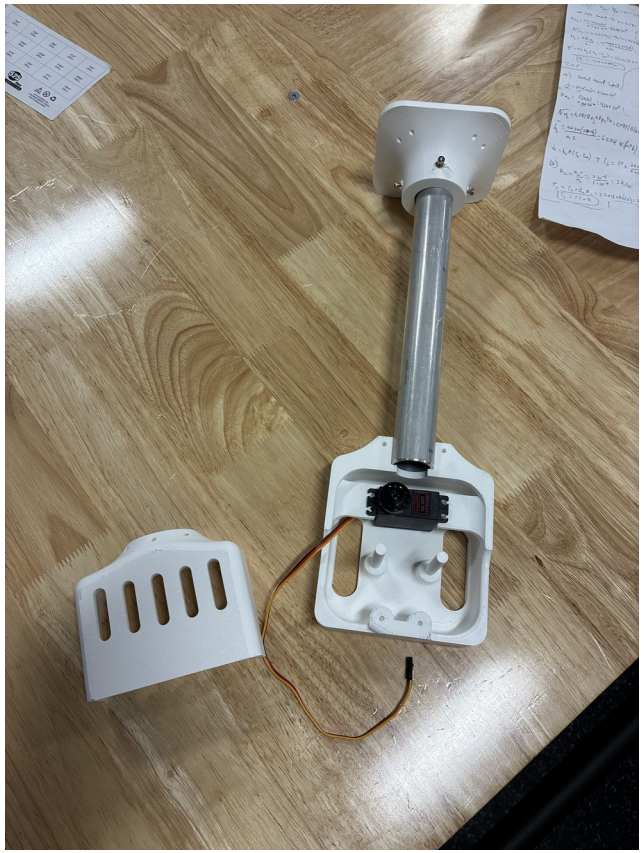
RoomCleaner

A cable robot that picks laundry off the floor

Jasper Buntinx
Mechanical Engineering, Texas A&M University
October 27, 2026





About Me

- **Education:** Mechanical Engineering, Texas A&M University · *[graduation: Month 20XX]*
- **EcoFil:** a filament recycler: shredder, dehydrator and extruder
- **Sculpture:** pieces for the Austin Creek Show; fiber-laser cutting and welding
- **Outside of class:** *[hobbies and interests]*
- **Why this project:** *[one line: why a laundry robot]*

[Photo of you building: EcoFil, a sculpture, or the workshop]

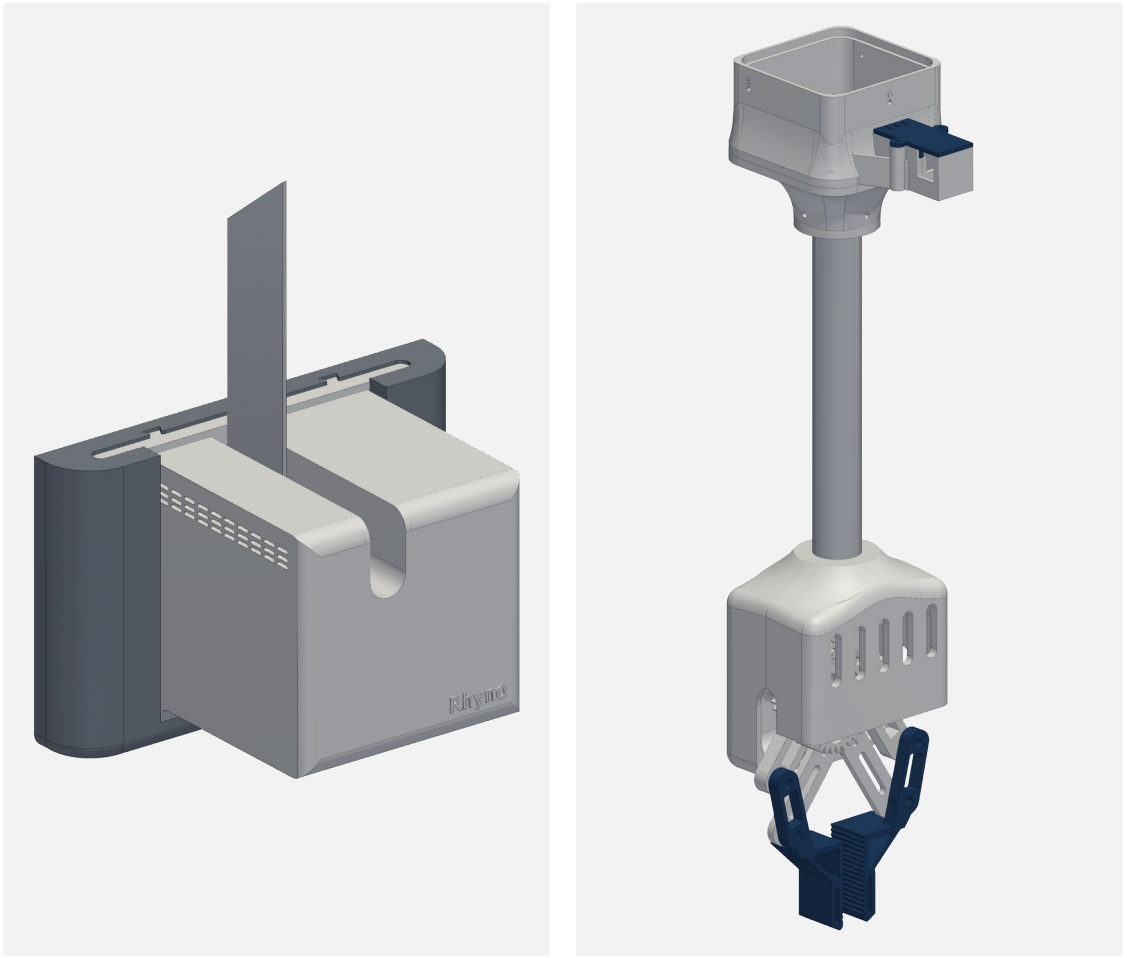
The Goal

- **Task:** laundry off the floor and into a hamper, anywhere in an 11 × 9 ft room
- **How:** four winches near the ceiling corners hold one claw on cables
- **Load:** jeans (0.9 kg) at 2 m/s²; first test: 1 lb
- **Limits:** 3D-printed PETG (polyethylene terephthalate glycol) plastic, hobby budget, safe in a home

What “done” means

Gate 1	Gate 2	Gate 3
One station runs repeatably	Stations hold and move 1 lb	One garment into the hamper

Now: gate 1 passed on the bench, gate 2 is this week’s test



Final design (CAD): a wall station and the claw

Scope and Timeline

Jul 16

Concept
Requirements, simulator, parts list

Design

Aug 6 – Sep 2

Gen 1
Corner plates, one central controller, finger claw

Design · Build · Test

Sep – Oct 6

Rethink
Sensors per winch, wiring, line outlets

Design

Oct 7 – 21

Gen 2 → final
SolidWorks: station v1→v3, spool v1→v3, claw → four-bar

Design · Build · Test

Oct 22 – 27

Build and test
Print, wire, three-station test

Build · Test

Me: a solo project (team of one)

- Requirements, calculations and test plans
- All CAD (computer-aided design) in SolidWorks: stations, spool, claw, fin
- 3D printing and fits; the heat fin on the shear and manual mill
- Wiring, soldering, electronics bring-up and debugging
- Control software: *[confirm the parts you wrote]*

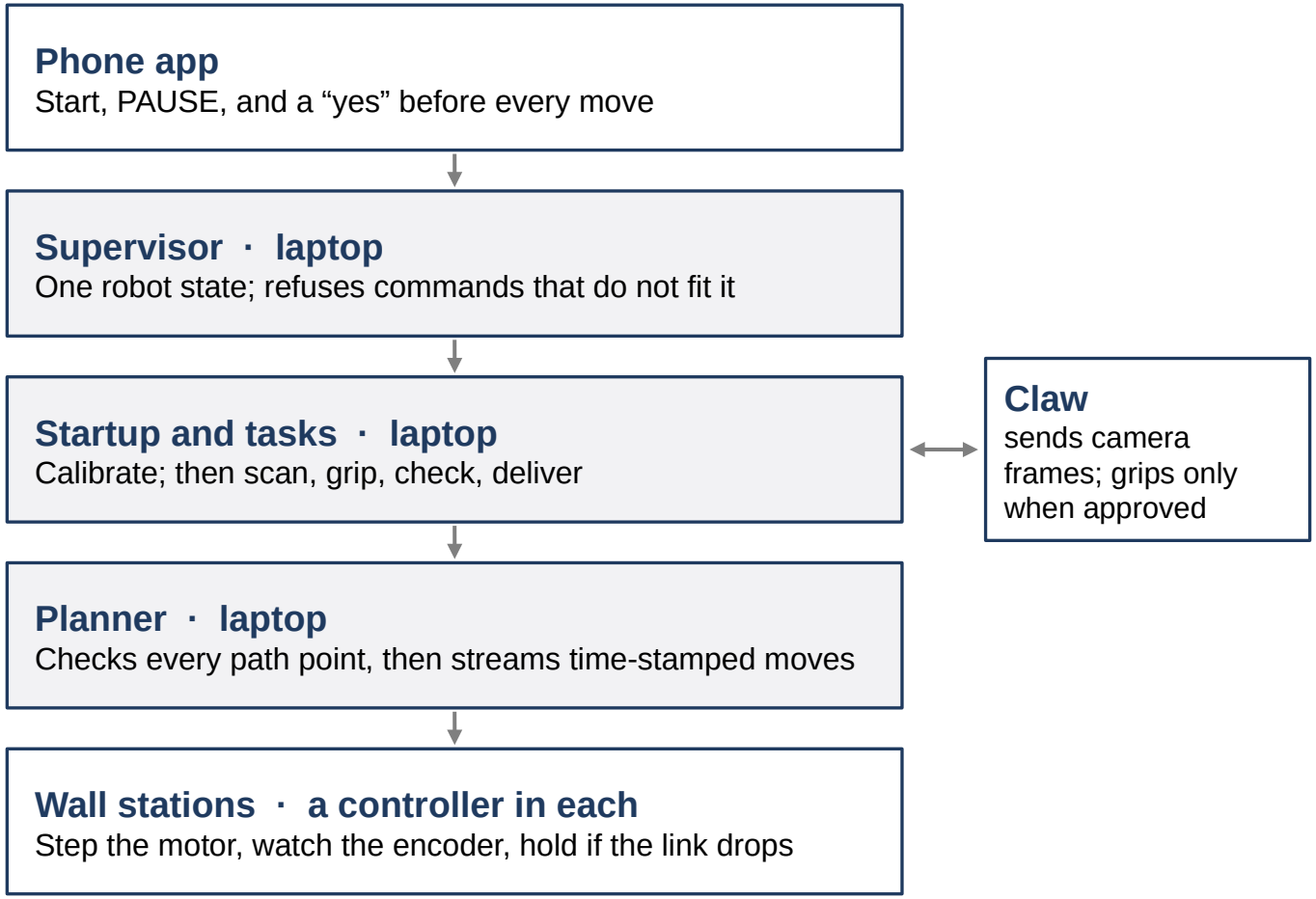
Bought or outside

- Stepper motors, drivers, controller boards
- Servo, camera, battery, encoder boards
- Garment detector: YOLO-World, an open-source model
- Help: *[shop access or mentors, if any]*

How It Works

Simulated end to end

- **Four wall stations:** winch, spool, encoder, heat fin, their own controller
- **One claw** on four lines: gripper, camera, battery
- **Laptop** plans every move and checks cable tension and obstacles
- **Phone:** start, PAUSE, and a “yes” before every move



Commands flow down; status and camera frames flow back up.

What I’m Optimizing

NOT MET

Lift capacity

Maximize

Target: Claw + jeans anywhere: 1.76 kg

Now: 1.56 kg at 1.5 A (calculated)

NOT MET

Claw mass

Minimize

Target: 0.45 kg budget

Now: ≈ 0.86 kg with battery (CAD); *[weighed: ___ g]*

TIGHT

Motor temperature

Limit

Target: Under 60 °C (PETG creep)

Now: ≈ 58 °C prototype fin; ≈ 45 °C with the 1/8 in fin (calculated)

MET IN DESIGN

Working area

Maximize

Target: The whole 11 × 9 ft floor

Now: All but 0.2 m at the walls (simulated)

MET IN DESIGN

Wiring

Minimize

Target: One plug per station

Now: One 12 V pair; 0.51 V drop on the longest run

TEST PENDING

Accuracy

Maximize

Target: ≤ 2 mm per cable over 300 mm

Now: Floor board 0.7 mm (calculated)

TEST PENDING

Speed

Maximize

Target: 50 mm/s, 2 m/s²

Now: In every tension check; not yet run

TEST PENDING

Cost

Minimize

Target: Hobby budget; parts I can buy and fix

Now: *[total spent: $___]*

Not traded: safety. Anchors in studs, one hard stop, holds on any fault, nobody under the robot

Trade-offs

Motor current ↑

Gain: pull, T ∝ I (+41 % at 1.5 A)

Cost: heat, P = I²R (2.0 → 4.1 W)

Set: 1.5 A, vented shell + fin

Claw strength ↑

Gain: gear safety factor 0.4 → 2

Cost: mass; payload = capacity − claw

Set: module 1.5 gears; mass budget open

Room size ↑

Gain: more floor covered

Cost: longer wire, more resistance: R = ρL/A

Set: only 12 V travels, on 16-gauge wire

Reach toward the ceiling ↑

Gain: more working volume

Cost: tension, T = W / (4 sin φ): 12 → 32 N

Set: stop 0.4 m below the anchors

Spool radius ↓

Gain: pull, T = τ / r

Cost: line speed, v = ω r; less line per layer

Set: Ø15 core: 34 N empty, 25 N full

Safety margins ↑

Gain: no skipped steps, no wall contact

Cost: usable force (÷ 1.5) and floor area

Set: ÷ 1.5; 0.2 m walls, 0.4 m ceiling

How they chain:

Claw mass ↑ → **Pull needed ↑** → **Motor current ↑** → **Heat ↑ as I²** → **Fin and vents**

Cable Tension

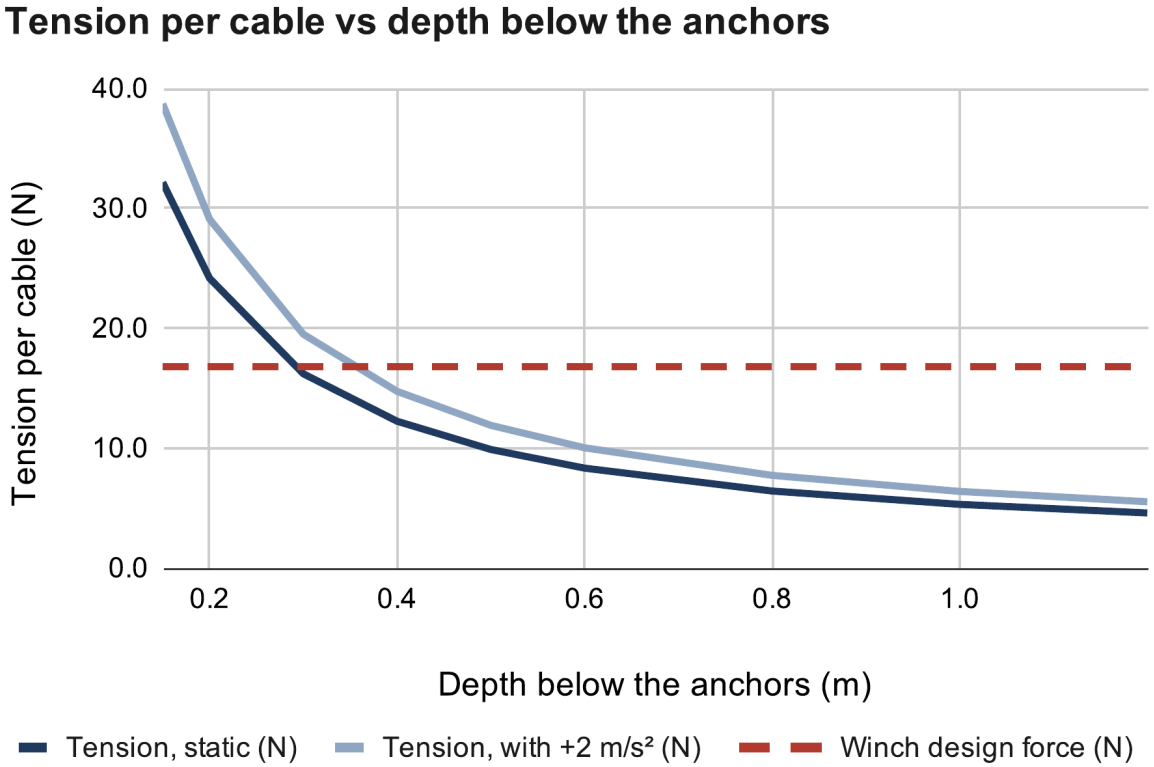
Calculated

- **Cables only pull:** every tension must stay above zero
- **Near the ceiling** the cables flatten and tension climbs
- 32 N per cable at 0.15 m below the anchors; 8.3 N at 0.6 m
- **Decision:** keep 0.4 m below the anchors; at 1.5 A that holds even while accelerating (14.7 N vs 16.8 N)

Key equations

$$\Sigma T_i \mathbf{u}_i + m(\mathbf{g} + \mathbf{a}) = \mathbf{0}, \quad T_i > 0$$

$$T = W / (4 \sin \varphi), \quad \sin \varphi = d / \sqrt{(d^2 + D_h^2)}$$



My calculation sheet (Google Sheets), “Cable tension” tab. Calculated, not measured.

Winch Pull

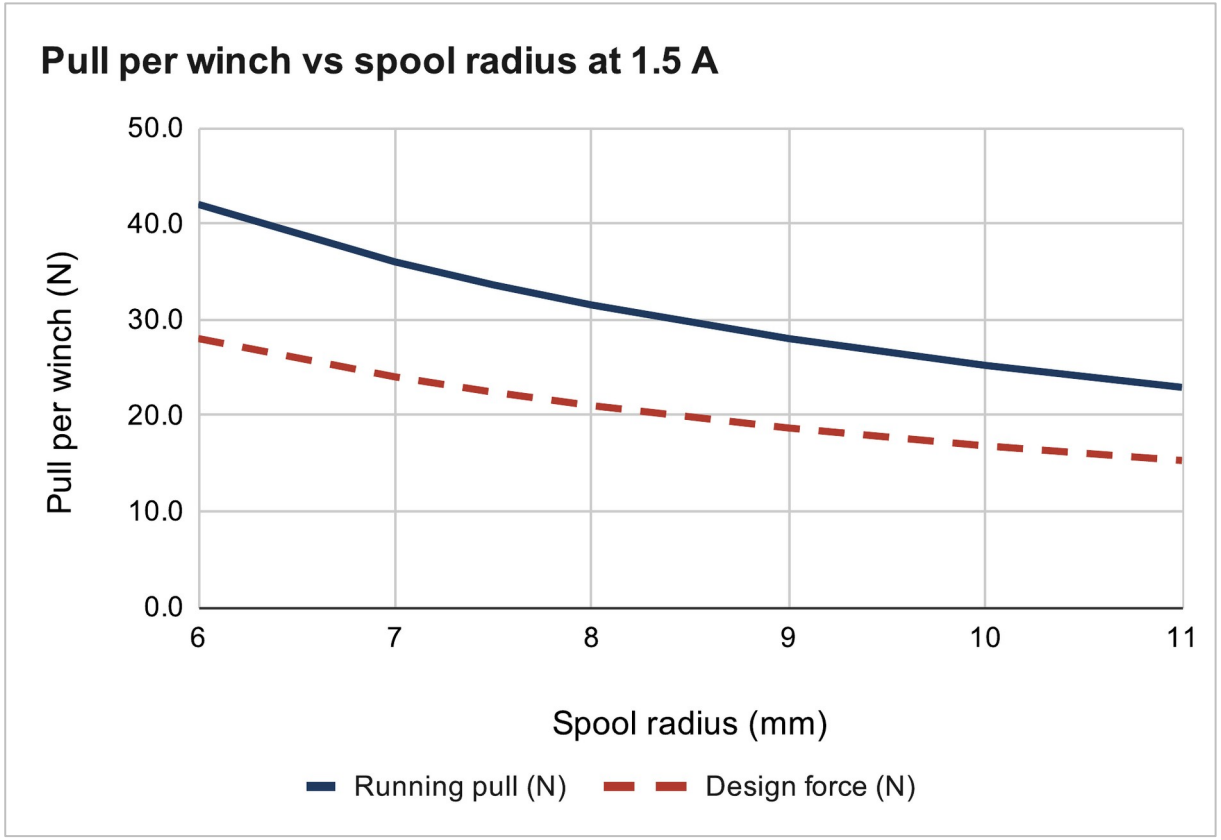
1.5 A set · pull calculated

- NEMA 17 stepper (standard 1.7 in frame), printed spool
- **Driver current raised to 1.5 A,** the motor's rating (was 1.06 A)
- Pull: 34 N empty spool, 25 N full
- **Design force 16.8 N** (was 11.9 N): full-spool pull ÷ 1.5 margin
- **Next:** a luggage-scale pull test replaces my torque assumption

Key equations

$$I = V_{ref} / (8\,R_s) = 1.2\ V / (8 \times 0.1\ \Omega) = 1.5\ A$$

$$T = \tau_{run} / r: \quad 33.6\ N \text{ at } r = 7.5\ mm, \quad 25.2\ N \text{ at } r = 10\ mm$$



My calculation sheet (Google Sheets), “Winch pull” tab. Calculated, not measured.

One Controller per Winch

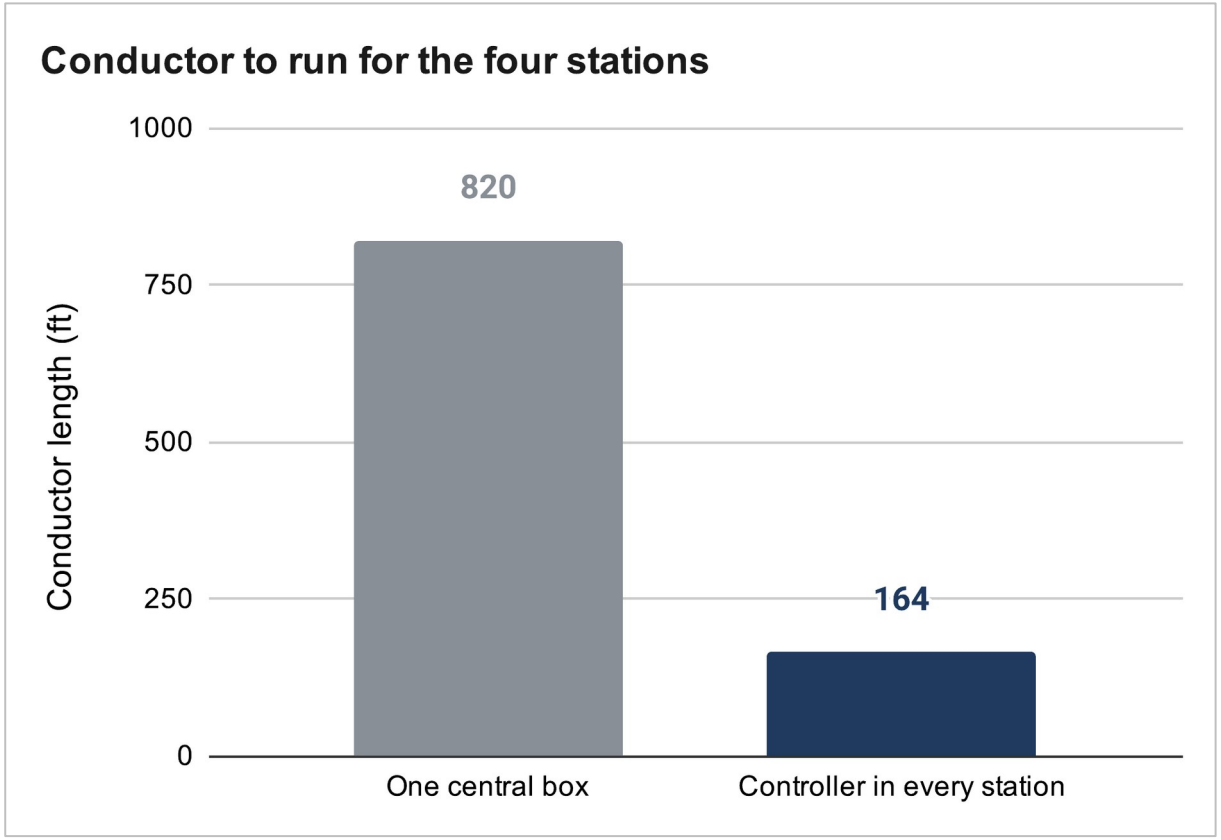
Built

- **Gen 1:** one board stepped all four motors; sync was free
- **Problem:** a switch and encoder per winch → 10 wires per corner
  - 820 ft of conductor; the sensor bus (I²C, Inter-Integrated Circuit) is meant for centimetres
- **Decision:** a controller in every station; only 12 V travels
- **Result:** 5× less wire; each station is built and tested alone

Key equations

$$\text{Conductor} = n \cdot \Sigma L: \quad 10 \times 82\ \text{ft} = 820\ \text{ft} \quad \text{vs} \quad 2 \times 82\ \text{ft} = 164\ \text{ft}$$

$$\Delta x = v \cdot \Delta t = 50\ \text{mm/s} \times 50\ \text{ms} = 2.5\ \text{mm}$$



My calculation sheet (Google Sheets), “Wiring” tab. Calculated, not measured.

Cooling the Motor

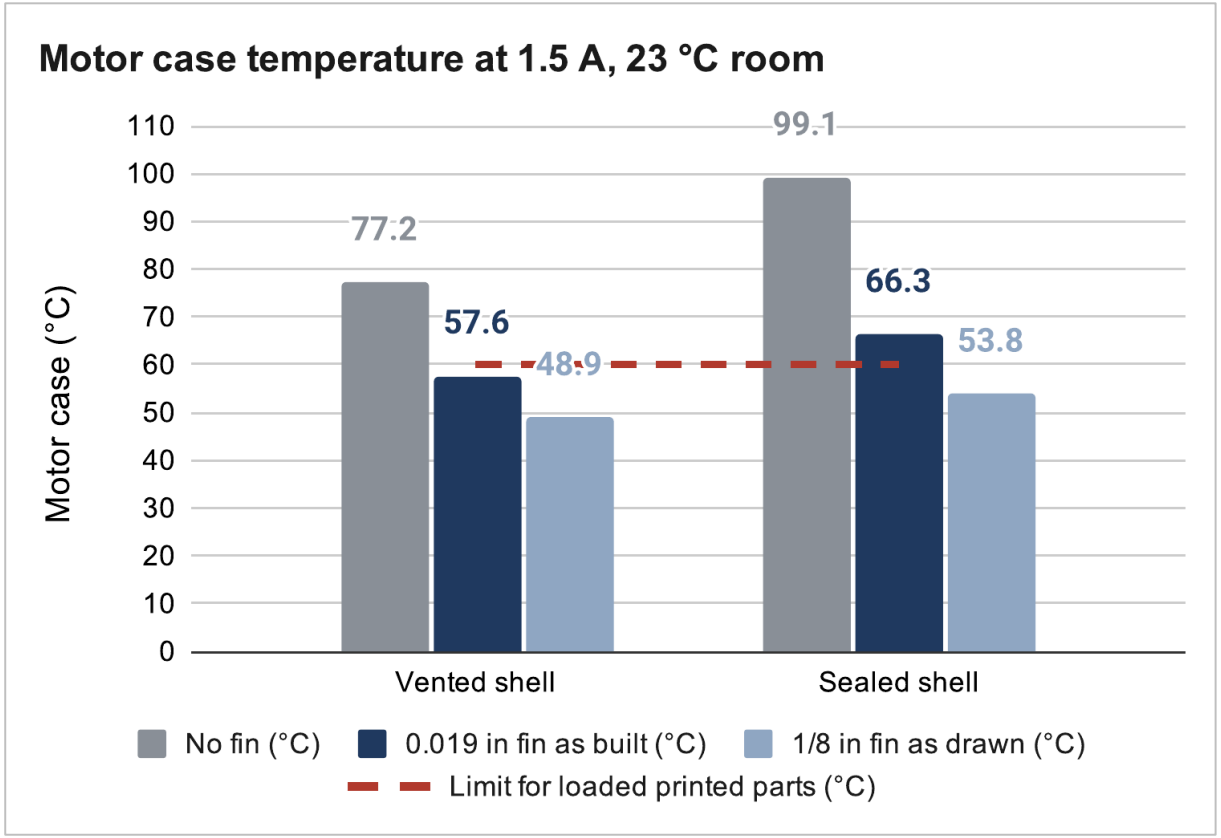
Fin built · temperature calculated

- **Problem:** a sealed printed shell traps motor heat; PETG creeps above 60 °C
- **Decision:** an aluminium fin up through the shell, shaped like an antelope horn
- **At 1.5 A:** twice the heat (4.1 W); prototype 0.019 in fin ≈ 58 °C, 1/8 in design fin ≈ 45 °C
- **Next:** 90-minute hold test at 1.5 A; back to 1.35 A if it runs hot

Key equations

$$P_m = I^2 R = (1.5\ \text{A})^2 \times 1.8\ \Omega = 4.1\ \text{W}$$

$$1/R = 1/R_{shell} + 1/R_{fin}, \quad \Delta T = P \cdot R$$



My calculation sheet (Google Sheets), “Motor temperature” tab. Calculated, not measured.

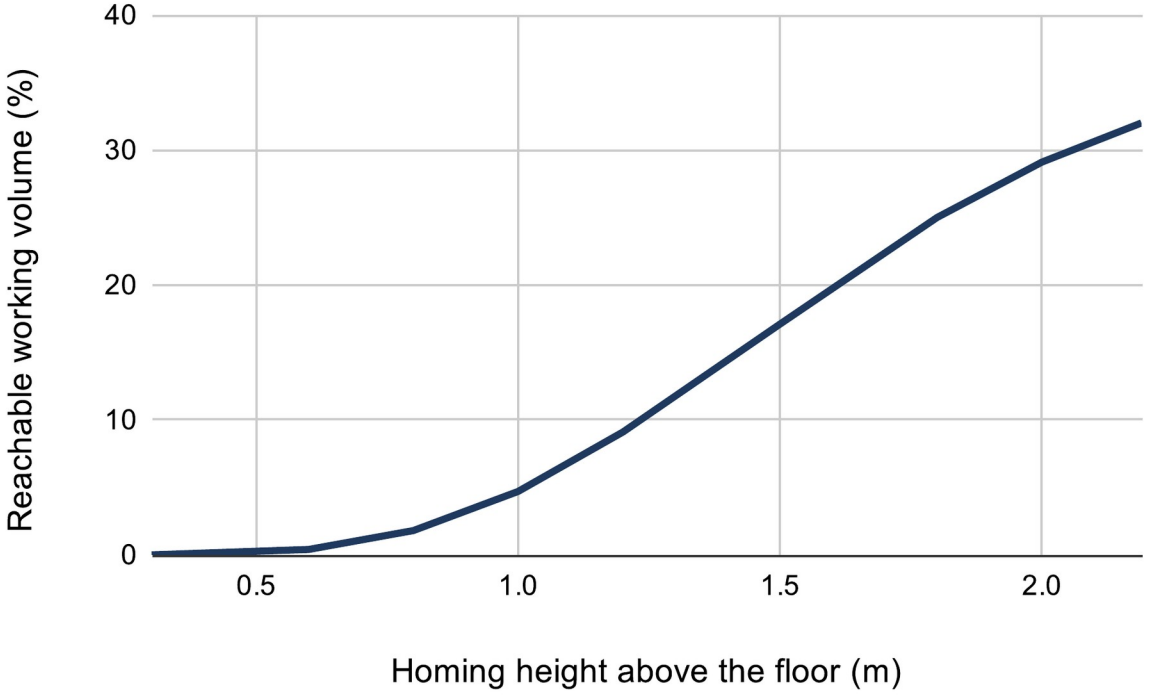
Homing

Calculated · simulated

- **Plan A:** a bead on each cable trips a switch → known length
- **Found:** a bead cannot pass the outlet, so cables can never be shorter
  - homing 1 m up leaves 5 % of the room reachable; 2 m up, 29 %
- **Decision:** the claw camera reads a printed floor board; fit the geometry from several poses
- **Result:** whole room reachable; the beads became end stops

Key equations

$$L_i(\mathbf{p}) \geq L_{i,\text{home}} = \|\mathbf{a}_i - \mathbf{p}_{\text{home}}\| \quad \text{for every cable } i$$



My calculation sheet (Google Sheets), “Bead homing” tab. Calculated, not measured.

The Claw

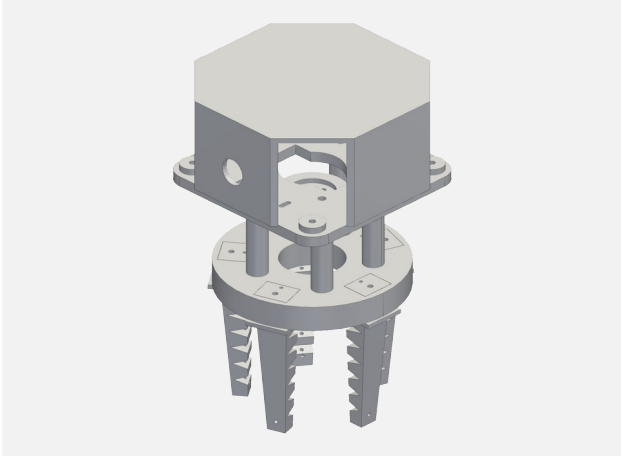
Body built · hands in CAD

- **Gen 1:** five tendon-driven fingers in TPU (flexible thermoplastic polyurethane)
- **Problems:** could not pinch thin fabric; one clamp was 4.6× over its mass budget
- **Final:** a geared four-bar, parallel jaws, 53 mm opening
- **Gears:** module 1 failed the stall check → module 1.5, safety factor ≈ 2

Key equations

Grip: $2\,\mu\,N \geq m\,g \;\rightarrow\; N \geq 7.4\ \text{N per jaw} \quad (1\ \text{lb}, \mu = 0.3)$

Lewis: $\sigma = F_t / (b\,m\,Y), \quad F_t = 2\tau / d$







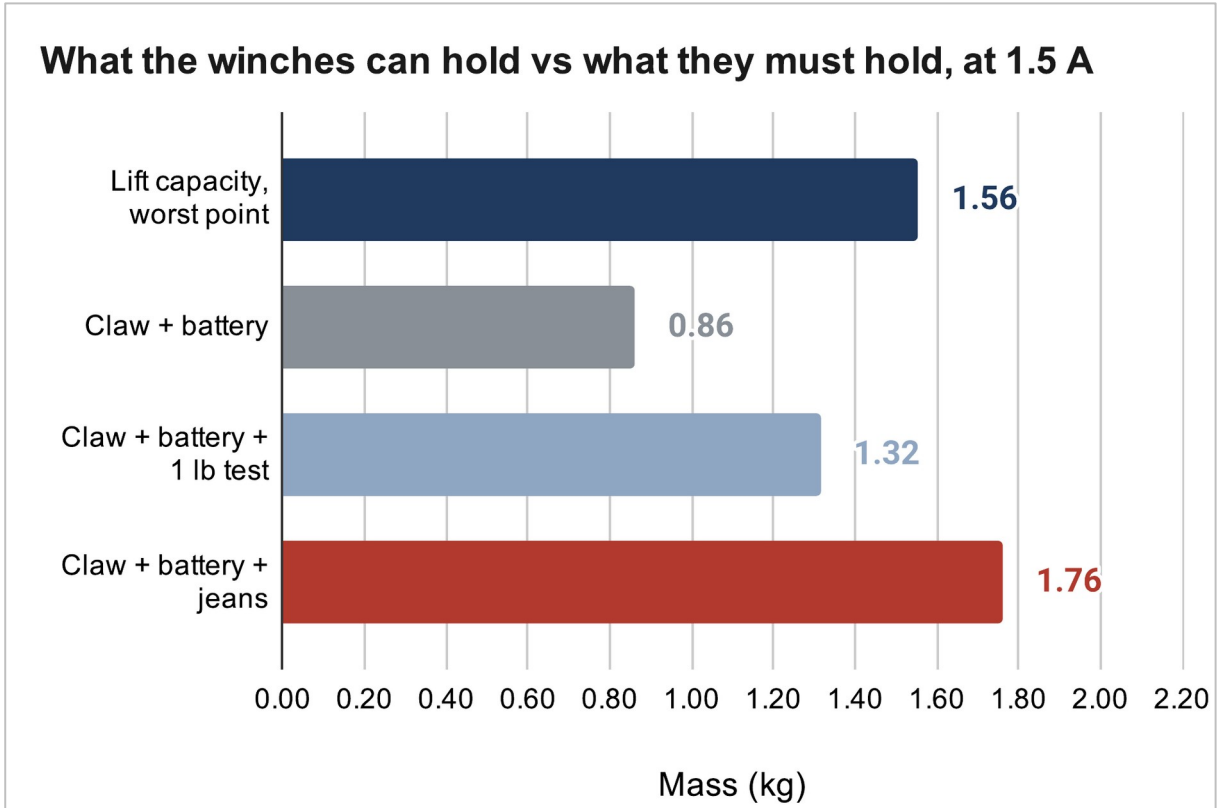


Fingers → v6 clamp → final geared four-bar (CAD) → printed gear train with the servo

Weight: the Open Problem

Open problem

- **Capacity at 1.5 A:** about 1.56 kg at the worst point (was 1.10)
- **Claw:** about 0.76 kg printed (CAD) + 0.1 kg battery; *[weighed: ___ g]*
- **1 lb test now fits** (1.32 kg); jeans are still 0.2 kg over
- **Options:** lighter claw · 3:1 winch reduction · lighter demo load
- **Next:** the pull test decides



My calculation sheet (Google Sheets), “Lift budget” tab. Calculated, not measured.

Key equations

$$\Sigma T_i u_{iz} = m(g + a), \quad 0 < T_i \le 16.8\ \text{N}$$

Printed Fits

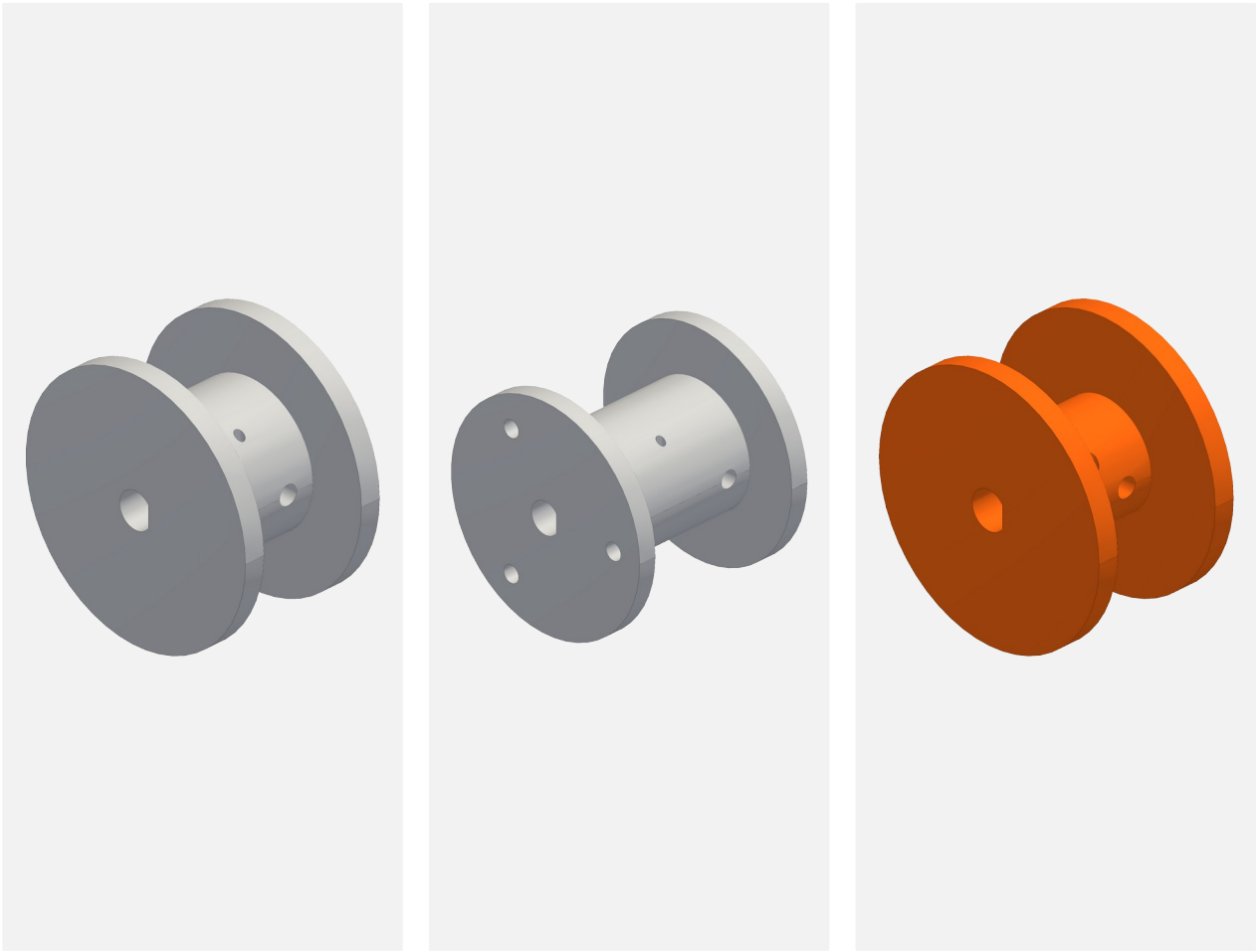
Built

- PETG on a Creality K1, 0.2 mm layers
- Printed holes come out undersize, so every fit gets an allowance
- **Spool:** v1 too long · v2 bore too tight · v3 Ø5.45 with a flat
- Torque goes through a set screw on the shaft flat, not the press fit
- **Lesson:** print a fit coupon before the part

Key equations

$$\text{Clearance } c = (D_{hole} - u) - D_{shaft}, \quad u = \text{printer undersize}$$

$$\text{Ø5.2 too tight} \rightarrow u \geq 0.2\ \text{mm, so Ø5.45 gives } c \approx 0.25\ \text{mm}$$



Spool v1, v2, v3 (CAD)

Machining the Fin

Built

- **Shop drawing first:** 1:1 print with a hole template
- Prototype: sheared 0.019 in sheet I had on hand; the design fin is 1/8 in
- **Manual mill, one setup:** five holes from the same two edges, ±0.1 mm
- Matte black paint; thermal paste on the motor face
- **Result:** bolted onto the motor with no reaming







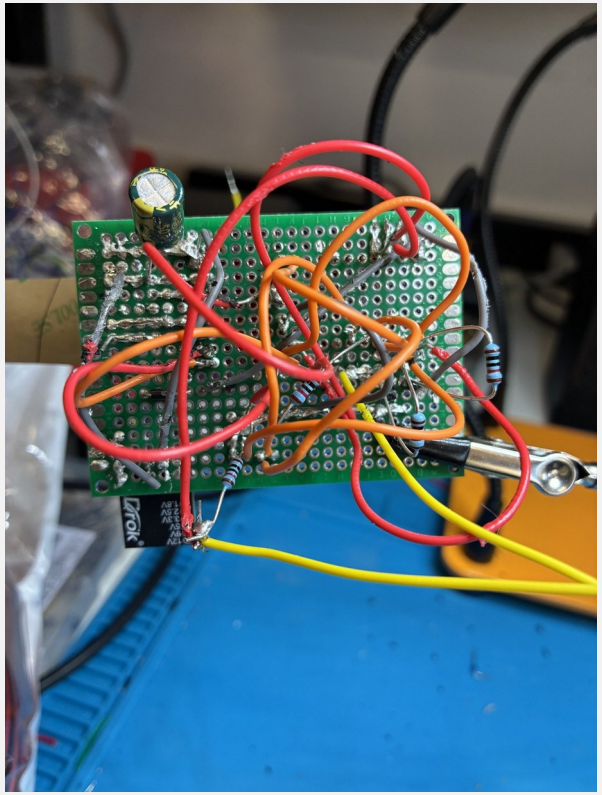


Shear and blanks · mill setup with the drawing · drilling · the finished fin

Electronics Bring-Up

Measured

- **Station board:** the driver boots off, so a reset never moves the motor
- Schematic built from the soldered board: 16 nets, 55 pins match
- **Failure:** the log said stepping; the shaft did not turn
- **Cause:** logic rail at 1.14 V, not 3.3 V: a bad connection
- **Now:** a meter checklist before any code





Soldered station board (left); bench bring-up (right)

First Test

Planned · this week

- **Three stations first:** the fewest cables that hold a point, so every tension is solved exactly
- **Sequence:** self-check → slack take-up → lift → calibrate → short move → pickup
- **Load:** 1 lb test weight, then one garment
- **Pass limits** written before the run

[Photo or video from the three-station test]

Check	Pass limit	Result
Startup self-check	All stations answer; slack taken up	*[fill in]*
Calibration fit	Held-out poses inside the safety margin	*[fill in]*
Short move, 1 lb	300 mm at 50 mm/s, ≤ 2 mm per cable	*[fill in]*
Pickup	One garment into the hamper	*[fill in]*

Thermal Check

Calculated · simulation next

- **Question:** does the PETG mount stay under 60 °C at 1.5 A?
- **Hand calc:** motor 56–59 °C (prototype fin), ≈ 45 °C (1/8 in fin)
- **ANSYS:** motor, fin, motor screws, inserts and PETG mount from my assembly; 4.05 W
- **Hold test:** 90 minutes at 1.5 A; thermocouples on the motor and the fin
- **If too hot:** the 1/8 in fin, or drop to 1.35 A (≈ 51 °C)

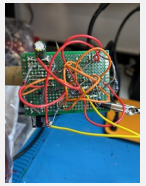
[ANSYS temperature plot: motor, fin, screws and inserts]

Method	Motor	PETG mount	Fin carries
Hand calc	56–59 °C	≈ 53–58 °C	1.4–1.6 W
ANSYS	*[fill in]*	*[fill in]*	*[fill in]*
Hold test	*[fill in]*	*[fill in]*	n/a

Three ways to know one temperature: calculate it, simulate it, measure it

Good and Poor Decisions

Good decisions



One controller per winch
5× less wire; each station tested alone



Slide-in wall mount
A station comes off; the anchor stays put



Camera calibration over bead homing
Whole room reachable; no tape measure

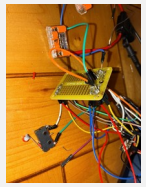


Gear strength checked at stall
Module 1 would have failed; fixed in CAD

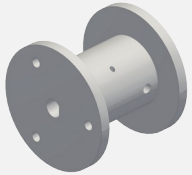
Poor decisions



Mass budget set late
Claw + jeans is 0.2 kg over, even at 1.5 A



Bead homing as plan A
Under a third of the room; a redesign



Spool printed before a fit coupon
Three prints to fit the shaft



Several stations before one was proven
Every fix repeated on each station

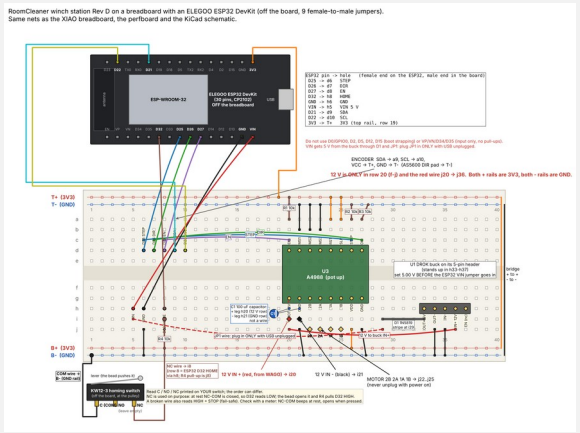
Thorough vs Scrappy

Thorough



- **Heat fin:** equations → shop drawing → one mill setup, ±0.1 mm
- **Station board:** schematic checked net by net against the solder
- **Gears:** strength checked at servo stall before printing

Scrappy



- **Fried a controller:** ESP32 on nine jumpers, same firmware
- **Floor board:** paper tags, 250 mm grid
- **Prototype fin:** 0.019 in sheet on hand; design is 1/8 in
- **Three stations** tested while the fourth is built

My rule: thorough where a mistake is expensive or unsafe; scrappy where it is cheap to undo

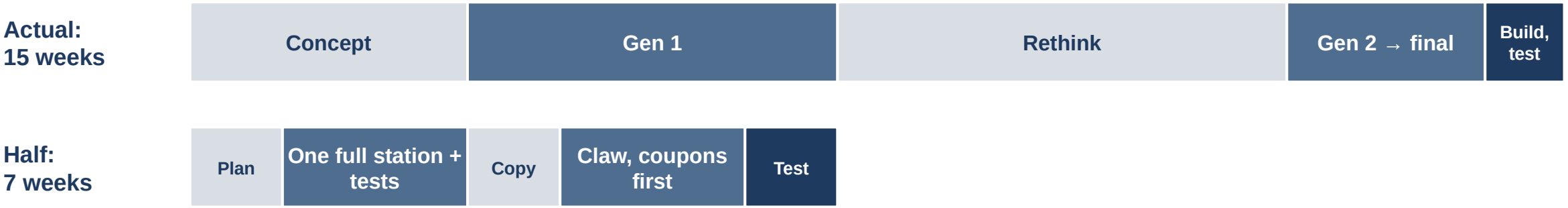
Lessons Learned

Do differently

- **Weigh early:** set the claw mass budget from the winch pull in week 1
- **Coupons first:** print the fit before the part
- **One station first:** prove it fully, then copy it
- **Meter before code:** check every supply rail first

In half the time

- **Freeze** requirements, mass budget and pass limits in week 1
- **Order long-lead parts** at once: pulleys, couplers, spare boards
- **Start with three stations:** fewest cables that hold a point
- **One generation per part,** each closed by a test



Same scale. Actual: Jul 16 to Oct 27. The half-time plan proves one station before copying it.

Questions

Jasper Buntinx · jasper_buntinx@tamu.edu

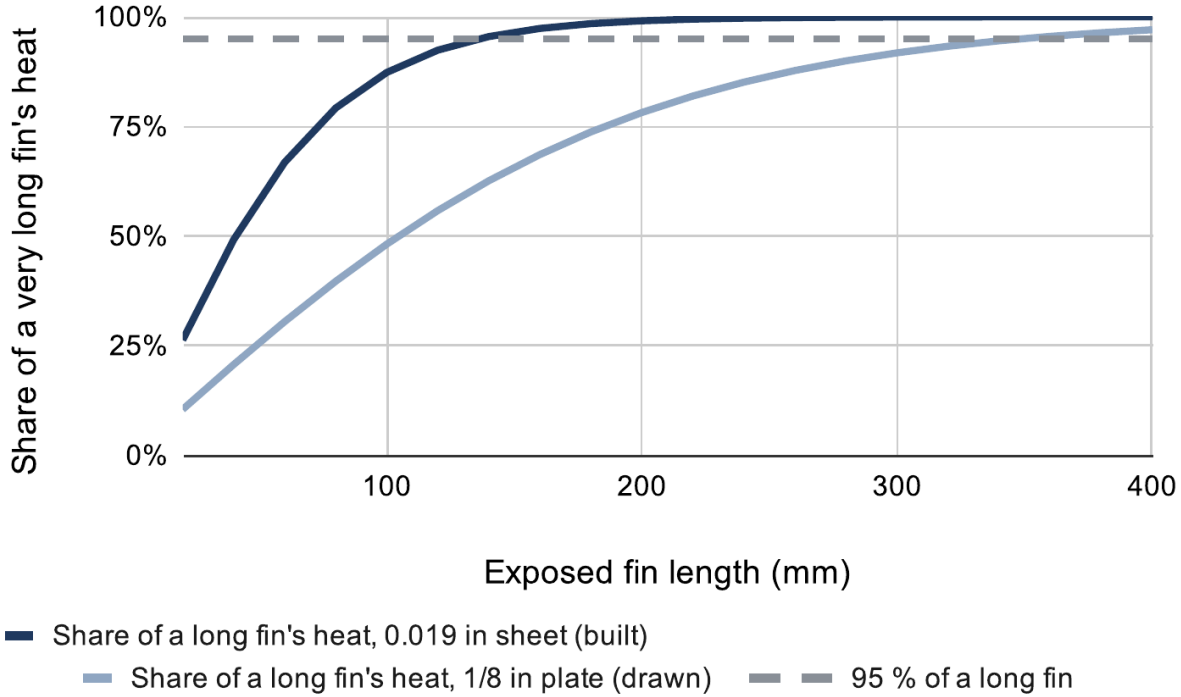
Where It Stands

Part	Status	Next test
Wall stations	Built; bench-commissioned *[how many]*	Three-station test, then the fourth
Station electronics	Built; drivers at 1.5 A *[how: Vref reading]*	90-minute hold test at 1.5 A
Heat fin	Prototype 0.019 in built; 1/8 in designed (D38)	Machine the 1/8 in fin; hold test
Thermal model of the station	Hand calc at 1.5 A; ANSYS model with screws set up	Run it; then the hold test
Claw body: tube, mount, servo, gears	Built	Weigh it
Claw hands, camera housing, encoder casing	Checked in CAD; printing	Fit checks on the first print
Calibration and control software	Simulated end to end	First hardware run
Winch pull	Calculated at 1.5 A: 25–34 N	Luggage-scale pull test

Fin Length

Calculated

- The far end runs cooler, so each extra millimetre does less
- **0.019 in sheet (built):** 95 % of the heat by 136 mm
- **1/8 in plate (first drawing):** 95 % by 349 mm
- Built 200 mm: past 136 mm the sheet adds under 5 %



Key equations

$$m = \sqrt{(2h / (k\,t))}, \quad \eta = \tanh(mL) / (mL)$$

$$k = 160\ W/m{\cdot}K,\ h = 7\ W/m^2{\cdot}K,\ t = 0.48\ mm \rightarrow L_{95} = 136\ mm$$

Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3

My calculation sheet (Google Sheets), “Heat fin” tab. Calculated, not measured.

Fin Sizing for 4 W

Calculated

- **Question:** what fin carries all 4.05 W alone, with the motor ≤ 60 °C?
- **Built 0.019 in sheet:** never; even infinitely long it carries ≈ 2 W
- **1/8 in (now the design), 200 mm:** 56–59 °C alone; ≈ 45 °C in the station
- **Thickness is the lever,** not length; bare needs 1/4 in × ≈ 300 mm

Key equations

$$q_\infty = w\sqrt{(2\,h\,k\,t)}\cdot\theta_b$$

$$m = \sqrt{(2h\,/\,(k\,t))},\quad \eta = \tanh(mL)\,/\,(mL)$$

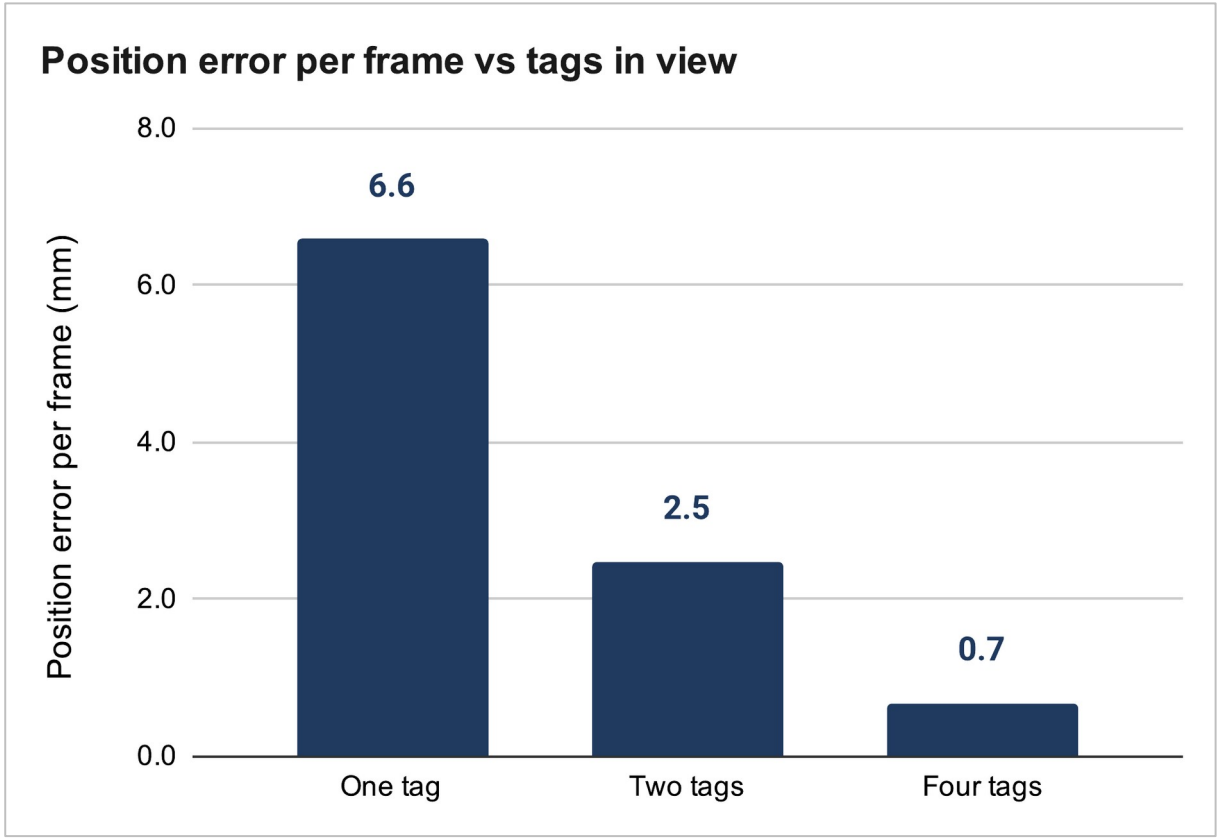
Painted 6061, 44 mm wide	Mass	Motor, 200 mm, fin alone	Shortest for ≤ 60 °C
0.019 in (prototype)	14 g	100–120 °C	never (max ≈ 2.0 W)
1/16 in	47 g	66–72 °C	never (max ≈ 3.6 W)
0.090 in	68 g	60–65 °C	210 mm or never
1/8 in (design, D38)	94 g	56–59 °C	145–185 mm
3/16 in	141 g	53–55 °C	125–140 mm
1/4 in	189 g	51–52 °C	115–125 mm

All 4.05 W through the fin, 23 °C room, paste contact; ranges: flange at one temperature to heat spread over the patch. Calculated.

Floor Board Accuracy

Calculated

- Angle error becomes position error through the lever arm
- One tag: 12.9 mrad × 0.51 m ≈ 6.6 mm
- Four tags in view: about 0.7 mm
- Averaging frames removes noise, not a fixed bias



Key equations

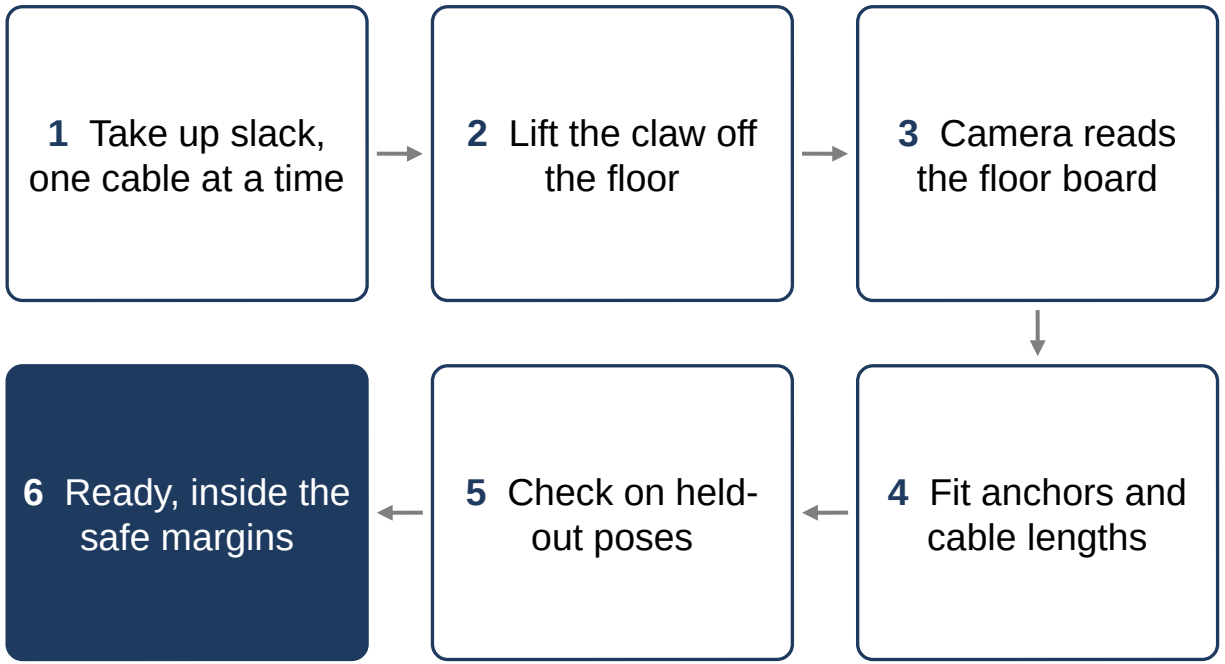
$$\delta x = \sigma_\theta \cdot L = 12.9\ \text{mrad} \times 0.51\ \text{m} = 6.6\ \text{mm}$$

My calculation sheet (Google Sheets), “Floor board” tab. Calculated, not measured.

Startup Calibration

Simulated

- No tape measure; nothing typed in
- Least-squares fit of anchor positions and cable zero lengths
- Held-out poses check the fit
- Full room only if the error bars fit inside the margin



Key equations

$$\min \Sigma_k \Sigma_i \left(L_{i0} + d_{ik} - \|\mathbf{a}_i - \mathbf{p}_k\| \right)^2$$

$$\text{Full room only if } k \cdot \sigma \le \text{margin}$$

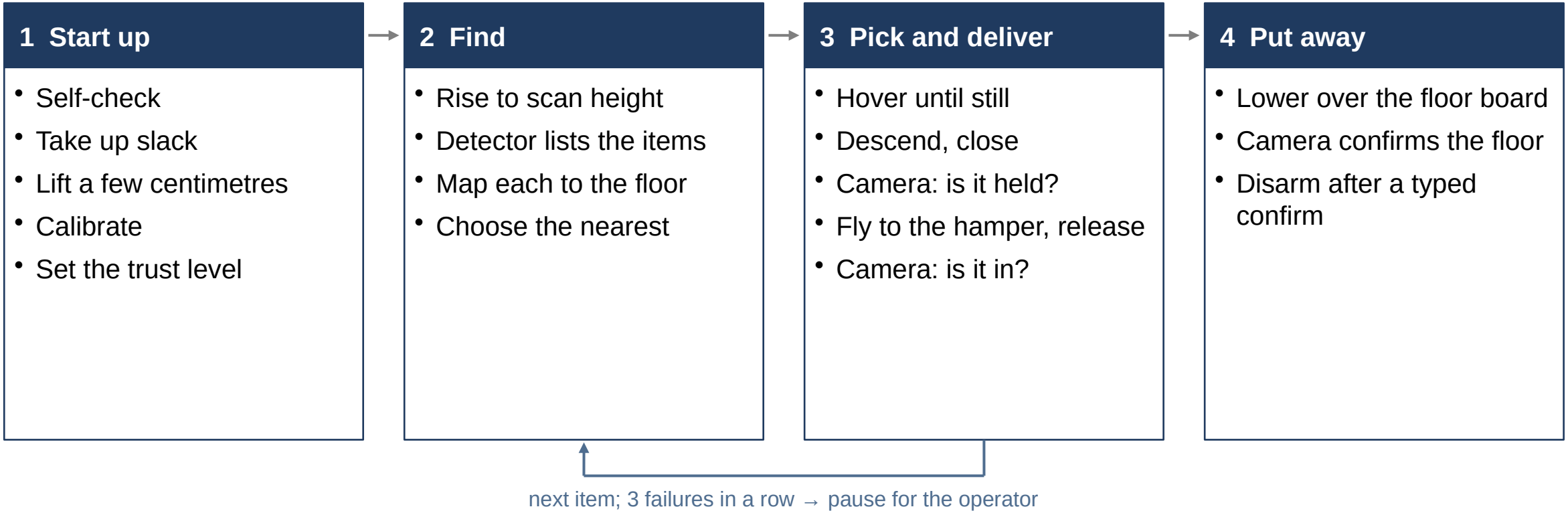
Logic: Gen 1 vs Final

Simulated

	Gen 1	Final
Where is the claw?	Assumed: where the cables were told to put it	Measured: claw camera on the floor board, refit at every startup
Room geometry	Point mass; anchors from a tape measure	Rigid claw; anchors and cable lengths fitted, with error bars
Who runs the motors	One central board steps all four	Each station runs time-stamped moves; the laptop coordinates
Finding laundry	Ceiling camera, pixel to floor map	Claw camera, ray from the claw's pose to the floor
Did the grab work?	Assumed once the claw closed	Camera checks pickup and delivery
Faults	Emergency stop only	Supervisor: a stop keeps torque; nothing restarts by itself

One Run

Simulated

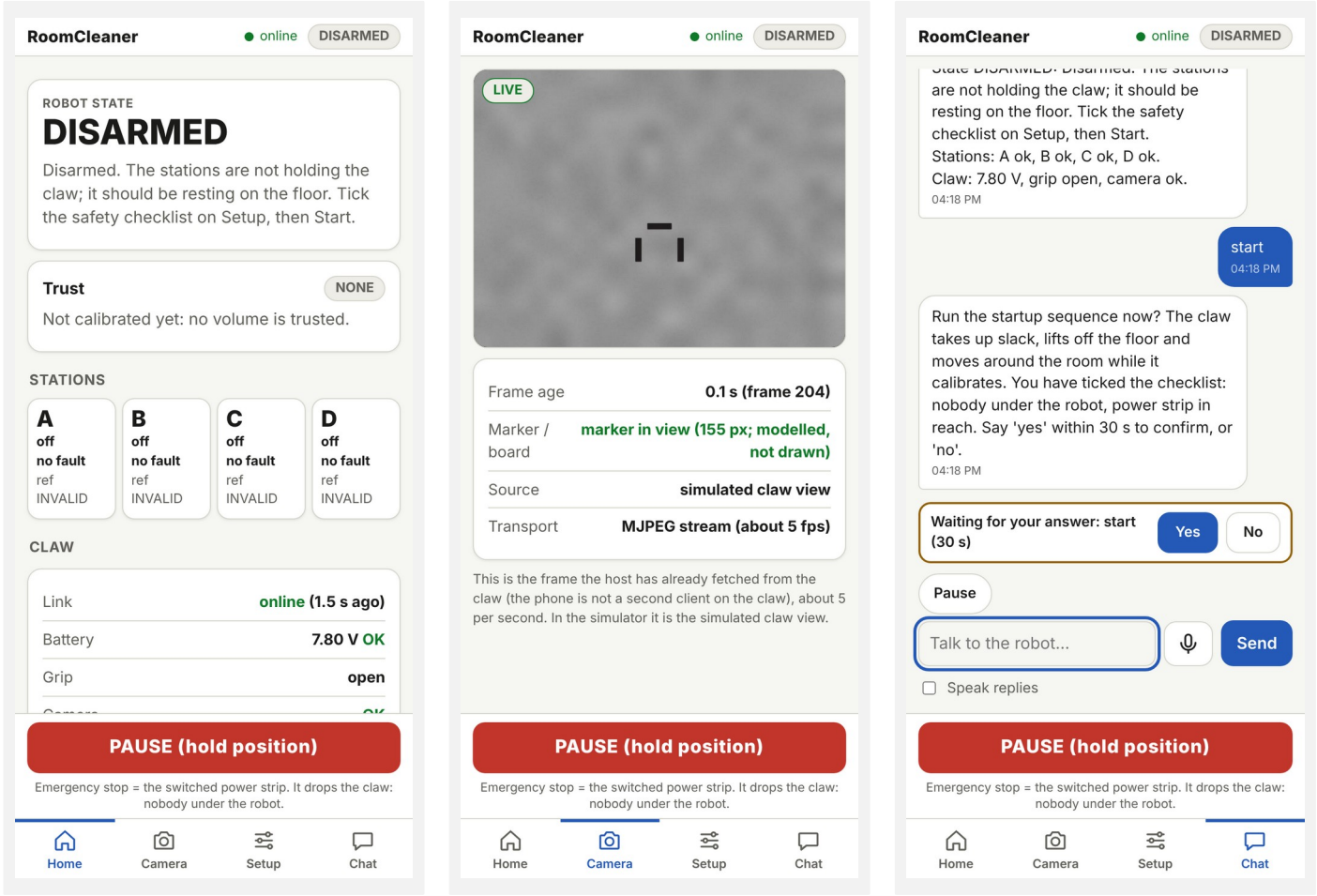


At any point: PAUSE or a fault stops every station on the same point of the path, still holding. Nothing restarts by itself.

Phone App

Simulated

- Status, live claw camera, guided setup
- Anything that moves asks “yes” first
- PAUSE holds position at once
- The switched power strip is the hard stop
- A reboot never resumes a task

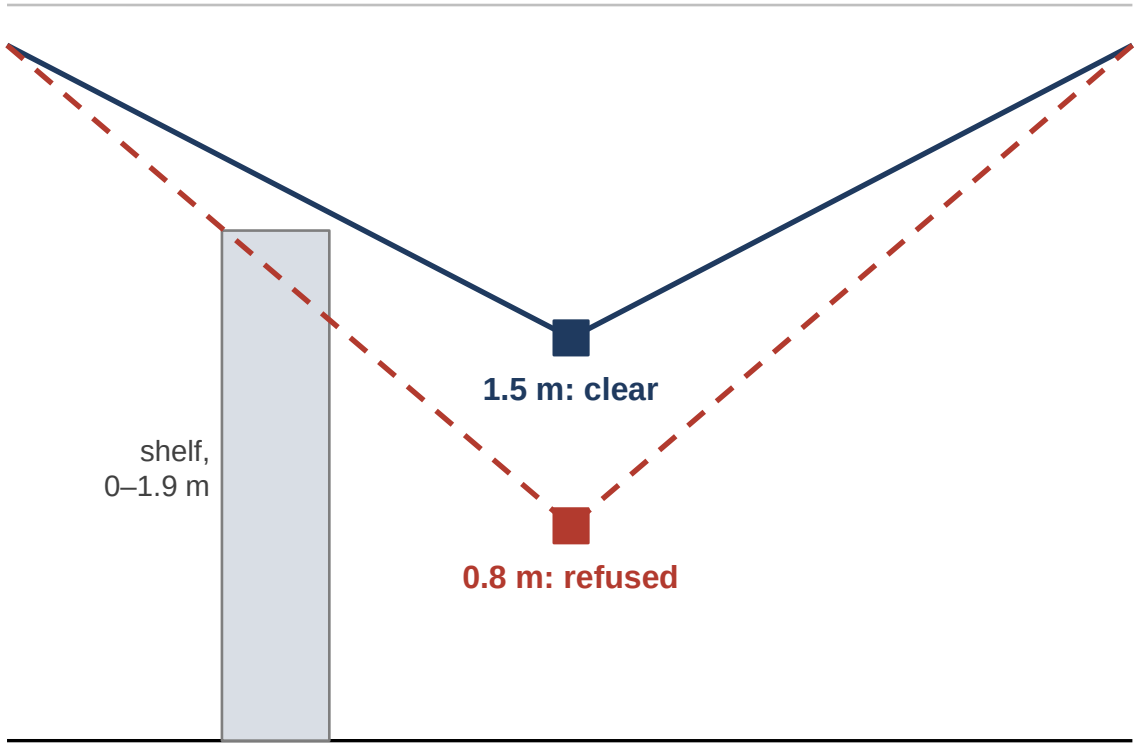


Phone app: home, camera, chat

Obstacles

Simulated

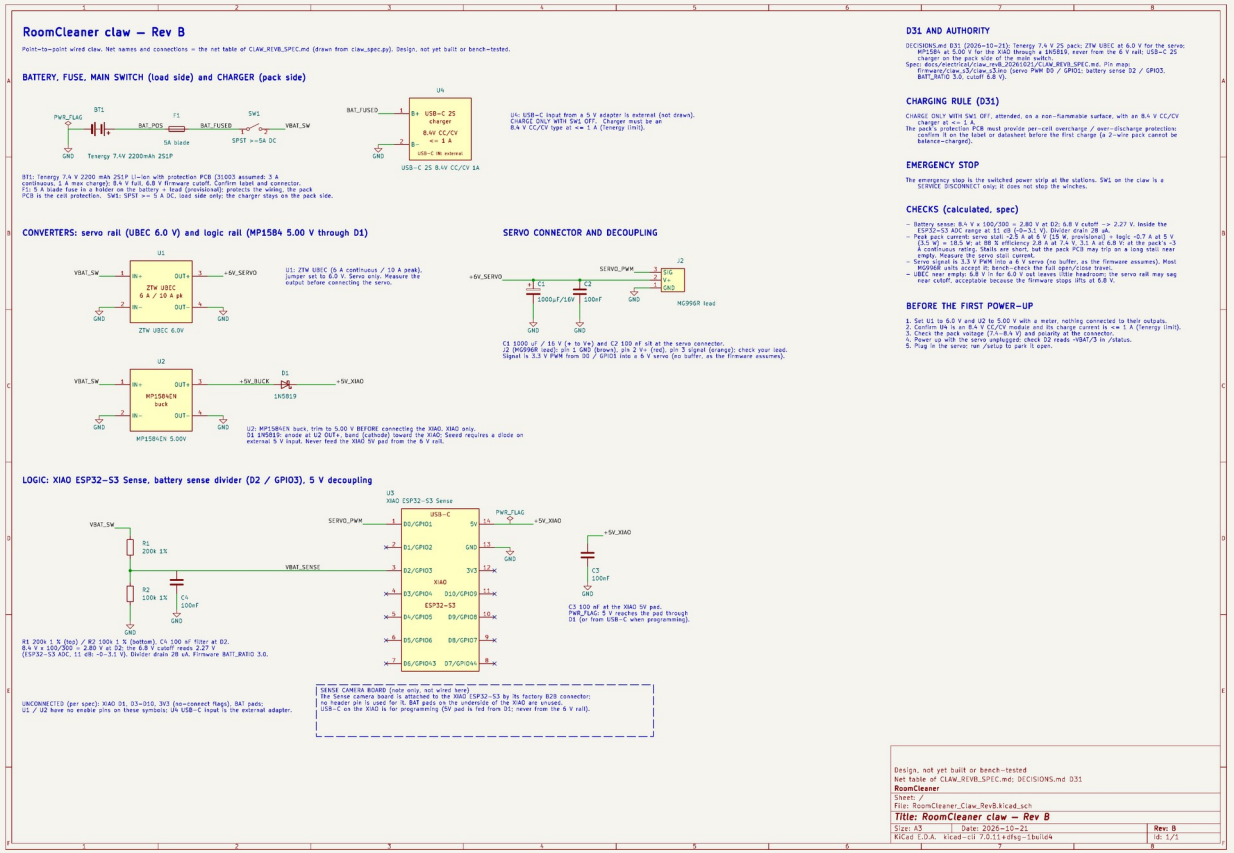
- Known obstacles are keep-out cylinders
- Every path point is checked: each cable against each cylinder
- A path that fails is refused before anything moves
- While moving: a stalled cable (encoder) or a failed grip (camera)
- Cannot yet see a person or a pet: nobody under the robot



Side view along a room diagonal: same floor spot, two heights; an example shelf

Claw Electrical

- One protected 2-cell lithium-polymer pack (7.4 V, 2200 mAh), 5 A fuse, main switch
- 6 V regulator for the servo only; separate 5 V for the controller
- Peak ≈ 2.8–3.1 A against a ≈ 3 A pack rating: stalls stay short
- Charged with the switch off, attended, 8.4 V at ≤ 1 A

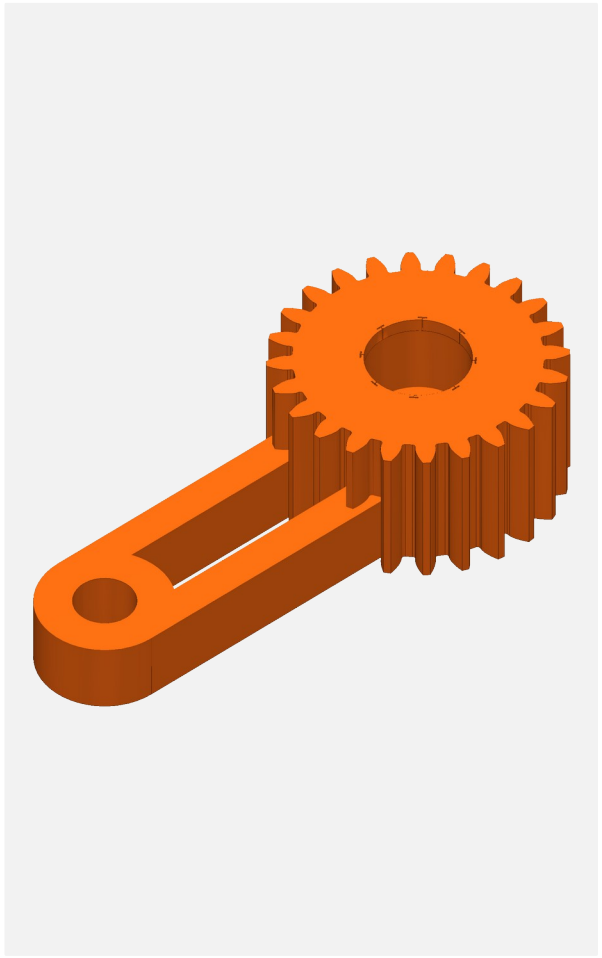


Gear Strength

Calculated

- Module 1.5, PETG allowable 22.5 MPa
- At servo stall: pinion safety factor 2.05, 36-tooth gear 2.50, sync gear 2.00
- Module 1 screened first: 0.35–0.41, so the module went up





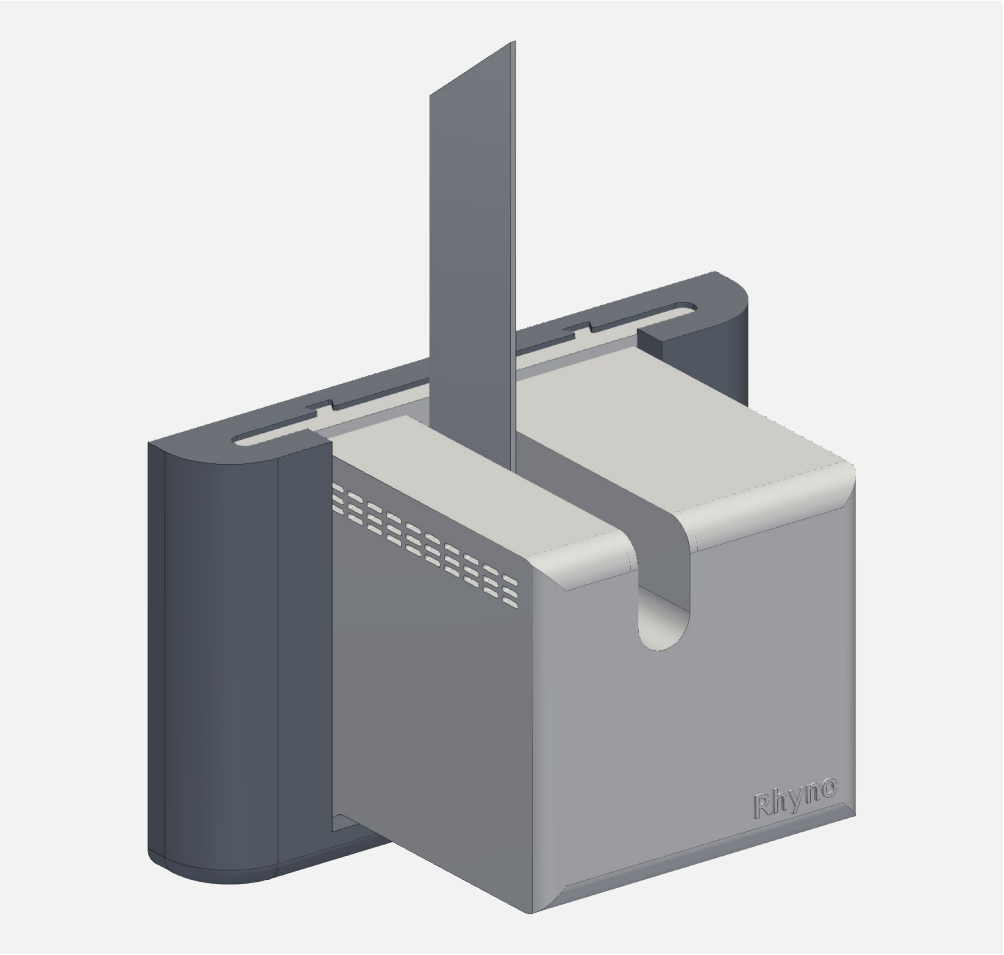
Key equations

$$\sigma = F_t / (b\, m\, Y),\quad F_t = 2\tau / d,\quad SF = \sigma_{allow} / \sigma$$

SF: safety factor; Y: Lewis form factor

Safety

- One hard stop: the switched power strip; faults and PAUSE hold torque
- Stations anchor into studs
- Nobody stands under the robot
- Never plug or unplug a motor under power
- No safety-rated function is claimed

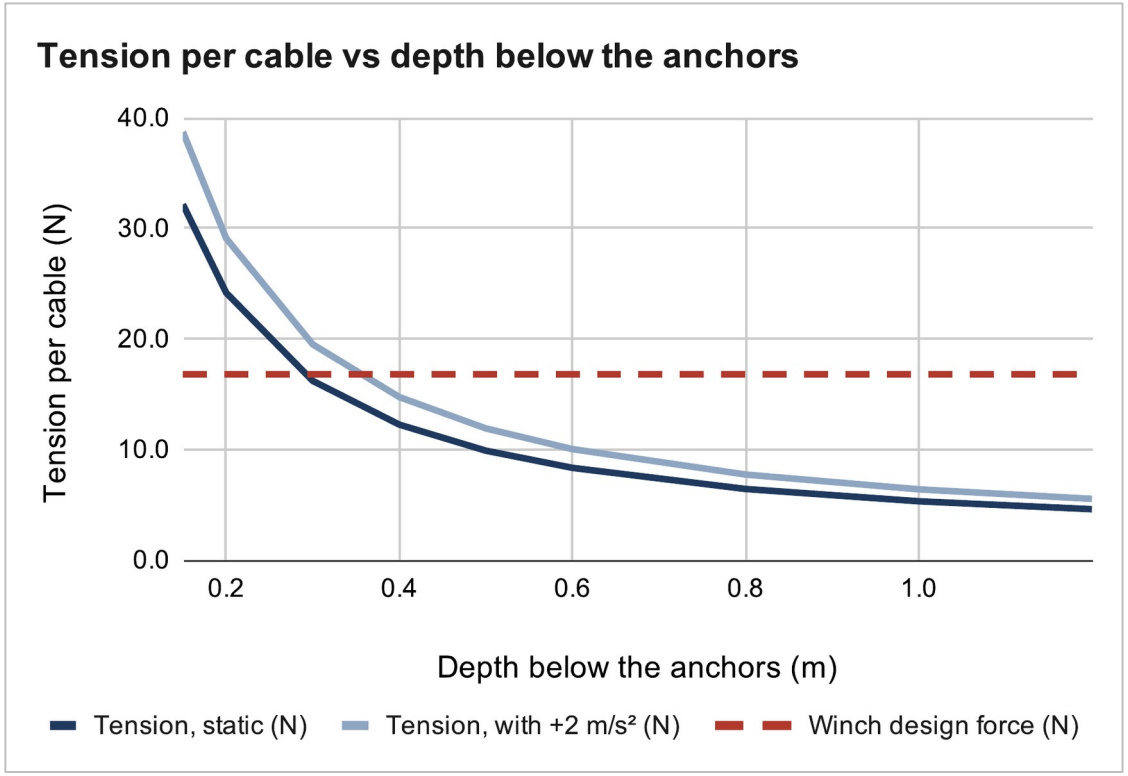


Acronyms

Term	Meaning
CAD	Computer-aided design
PETG	Polyethylene terephthalate glycol (the printed plastic)
TPU	Thermoplastic polyurethane (flexible printed plastic)
NEMA 17	National Electrical Manufacturers Association motor frame, 1.7 in square
I²C	Inter-Integrated Circuit: a short-range two-wire sensor bus
ESP32	A Wi-Fi microcontroller family (Espressif)
A4988 · AS5600	Stepper-motor driver chip (Allegro) · magnetic angle sensor (ams)
YOLO	“You Only Look Once”: an object-detection model family
SF	Safety factor: allowable stress ÷ actual stress

Calculations and Sources

- Every chart comes from my calculation sheet, one tab per topic
- Every result is a formula on marked inputs
- **Sources:**
  - Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3
  - Allegro MicroSystems, A4988 datasheet



My calculation sheet (Google Sheets), “Cable tension” tab. Calculated, not measured.

RoomCleaner engineering calculations (Google Sheets)